

Andrew Van Ostrand

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Education

Lehigh University

Expected December 2026

Bachelor of Science in Computer Science & Business (GPA: 3.67 / 4.00)

Bethlehem, PA

- **Relevant Coursework:** Data Structures and Algorithms (Java), Systems Software (C, C++), Foundations of Discrete Structures and Algorithms, Computer Architecture (ASM, C), Programming Languages, Database Systems (SQL, Java)

Projects

Truth Trail | Blockchain (SKALE), GCP, Gemini, Vertex AI, Solidity, Hardhat

Mar 2025

- Developed "Truth Trail," a full-stack evidence management system using blockchain (SKALE) for a tamper-proof chain of custody, Google Cloud for secure storage, and AI (Gemini Flash/Vertex AI) for automated image analysis.
- Engineered a dual-layer verification system (database and blockchain) and built both public verification and secure admin interfaces with search functionality to ensure evidence integrity and accessibility for stakeholders.

Prof Pair | AWS Comprehend, Amazon SageMaker, AWS S3, Flask, HTML

Oct 2024

- Built an AI-powered platform to match students with professors using sentiment analysis of course reviews.
- Prototyped for 8,500+ students at Lehigh and Lafayette using AWS Comprehend and SageMaker.
- Awarded "Best Use of AI."

LCRS | Google Cloud Vertex AI, AutoML, GCP Compute, Python

Mar 2024

- Developed a predictive model with Vertex AI/AutoML to assess lung cancer risk from medical history data.
- Built a cost-effective pre-screening tool to improve accessibility in low-income communities.
- Showcased potential for improved early detection via a user-friendly interface; awarded *Best First-Year Solution*.

ASL for the Community | AWS, S3, Polly, SageMaker, Python

Nov 2023

- Developed an image recognition model using Amazon SageMaker to translate ASL signs into text and speech, enhancing communication for the deaf community.
- Overcame challenges in model training due to time constraints, focusing on key ASL signs (A, B, C) for initial implementation.
- Awarded for Best AWS Integration: Recognized for exceptional use and integration of Amazon Web Services in the project.

Experience

TA - CSE262: Programming Languages

Aug 2025 – Present

Lehigh University

Bethlehem, PA

- Guided 90 students through core programming language concepts including recursion, memory management, and object-oriented design in Java by providing support in office hours and on Piazza.
- Assisted students in developing a Scheme subset interpreter (tokenizer, parser, and tree-walk interpreter), reinforcing understanding of syntax, semantics, and functional programming paradigms.

Research - Prof. Michael Spear

May 2025 – Present

Lehigh University

Bethlehem, PA

- Developing LLVM-based compiler framework to optimize Non-Volatile Memory (NVM) and RDMA access
- Implementing automatic code annotation propagation, reducing programmer complexity

Grader - CSE017: Programming and Data Structures

Oct 2024 – Present

Lehigh University

Bethlehem, PA

- Graded assignments and provide constructive feedback for 150+ students
- Mentor students in understanding fundamental programming, data structures, and algorithms

Technical Skills

Languages: Java, C, C++, Python

Technologies: GCP, Google Cloud AutoML, Google Cloud Compute, AWS Sagemaker, Amazon Polly, Amazon S3, Docker

Concepts: Compiler, Operating System, Artificial Intelligence, Machine Learning, API, Cloud Computing, Problem Solving, Analytical Thinking, Communication, Adaptability